PROCUREMENT STRATEGY

1. How will NASA assess "commercial viability" and what specific metrics will providers need to demonstrate? NASA plans to provide more information at an industry forum and/or in the Draft RFP regarding what specifics are required to provide proof of commercial viability; for instance, NASA plans to evaluate each provider's ability to show viability in addition to a government customer.

2. When assessing "commercial viability," will NASA consider the incremental actual cost basis of each vendor to deliver the initial and full capability? For example, a vertically integrated CLD partner which has already made significant R&D investments in subsystems, primary structure manufacturing, etc. will have lower incremental costs than a vendor using a traditional outsourced aerospace supply chain. As a result, a vertically integrated partner would then only require a more conservative and realistic revenue target overall to break even and be profitable, thereby making it more "commercially viable." NASA plans to provide more information at an industry forum and/or in the Draft RFP regarding NASA's evaluation criteria and what specifics offerors are required to provide to demonstrate proof of commercial viability.

3. Will NASA consider Funded Space Act Agreements for the development portion of Phase 2 leading to certification, instead of contracts? And then use contracts under the FAR for services? NASA does not plan to compete and award additional Funded Space Act Agreements at this time for the development of Commercial LEO Destinations Contract (CLDC). The agency currently plans for CLD Contract (Phase 2) to be competed and awarded as a FAR-based contract.

4. How will NASA equitably account for any private capital spent on hardware development, test, and demonstration related to a CLD bid, particularly if such capital investment drastically outweighs any funding received by that bidder from NASA before the bid? NASA plans to provide more information at an industry forum and/or in the Draft RFP regarding NASA's evaluation criteria. NASA may consider the investments that providers have made, impact to financial capability, and/or ability to close a business case.

5. Does NASA anticipate a FAR part 15 competition or FAR Part 35.016 competition or a multiple-award IDIQ type contract? NASA plans to provide more detailed information at an industry forum and in the Draft RFP regarding contract type. NASA is currently anticipating a FAR part-15 based contract, which allows for and may include IDIQ Task Orders, and, depending on available federal funding and receiving proposals of adequate merit, plans to award none, one, or multiple awards.

6. How many awards does NASA anticipate making as a result of the CLD Phase 2 competition? (We understand based on the slides that NASA will also provide for on-ramp opportunities.) NASA may award none, one, or multiple awards depending on available federal funding and receiving proposals of adequate merit. As noted, NASA also anticipates including an On-Ramp clause in the solicitation. More details will be released at an industry forum and/or in the Draft RFP.

7. How will NASA account for the delays experienced by the CDISS contractor when evaluating their proposed approach? NASA plans to evaluate Past Performance for all offerors.

8. How have the changes made to the CDISS habitat modules impacted the CLD Phase 2 acquisition strategy? Individual CLD providers' development during Phase 1 has not impacted the CLD Contract (Phase 2) acquisition strategy.

9. How has NASA leveraged lessons learned on the CDISS contract and CLD Phase 1 Space Act Agreements when developing the CLD Phase 2 acquisition strategy and will NASA share those lessons with all competitors? NASA is leveraging lessons learned from contracts and SAAs, along with other lessons learned across the entire agency and Government. NASA does not plan to share detailed acquisition lessons learned with competitors.

10. How have NASA's deorbit plans for the ISS impacted, if at all, the CLD Phase 2 acquisition strategy? NASA's desire to reduce or eliminate the gap between the CLD and ISS transition has influenced NASA defining a minimum initial operating capability by 2029 and may drive NASA's ISS end of life plans.

11. Does NASA think it is important (and how would you approach) having an equitable competitive field between companies without funded SAAs and those with funded SAAs (and the CDISS contractor if permitted to bid)? Will NASA include or take into account the previous funding and support NASA has provided some contractors when evaluating and comparing the CLD Phase 2 proposals? Prior NASA agreements and/or contracts have been competed as full and open competitions. NASA plans for the CLD Contract (Phase 2) procurement also to be conducted as a full and open competition. Competitors will be evaluated against the same criteria for award.

12. Current international partner cooperative efforts are embodied in international treaties and implemented through government-to-government barter agreements. Is NASA considering how to preserve these relationships under a commercial operations model? NASA plans to continue international relationships and build new ones in the future, and leverage them where possible, to address liability and other concerns associated with CLDs. NASA also intends to enter into Gov-Gov agreements around joint science and possibly other areas of cooperation in the CLD era. The nature of these agreements will depend on the areas of collaboration. NASA plans to request industry feedback at an industry forum and/or in the Draft RFP.

13. Will terms be similar to Commercial Crew Program or will there be a set of specific FAR based terms applied for this procurement? NASA is still evaluating the terms and conditions to be included in the CLD Phase 2 solicitation and resulting contract. Clauses are expected to be similar to Commercial Crew Program, however; NASA is evaluating lessons learned and contract clauses from all past contracts and potentially unique clauses associated with this procurement. NASA is currently planning to award a FAR-based contract.

14. When will the Source Evaluation Board be named/established? NASA will appoint the Source Evaluation Board members in the approximate timeframe when the Draft RFP is released per NASA's standard procurement process.

15. How will NASA deal with limiting CLD operator liability to a reasonable extent? NASA is evaluating options for a liability framework for the Phase 2 CLD Contract and will share more information for feedback at an upcoming industry forum and/or in the Draft RFP.

16. In the baseline acquisition strategy slide, what does NASA mean by "overlap between CLD and ISS capabilities" is desired? NASA is evaluating different mission scenarios and timelines to decrease the risk of a gap in human presence in LEO. NASA's goal is to not have a gap in continuous human presence and desires to have an operational CLD before deorbiting ISS at the end of the decade.

17. "Long term goal for NASA is to become one of many customers for commercial transportation operations in LEO…"
    a. **How will operational costs—such as crew transportation, life support systems, and maintenance—be shared between NASA and providers?** NASA is evaluating and developing its acquisition strategy with respect to cost structure and will provide more information at a future industry forum and/or in the Draft RFP.
    b. **Will NASA's required presence during key operations (e.g., launch, docking, emergency response) incur additional costs or delays for providers?** NASA is evaluating and developing its acquisition strategy with respect to services the

agency may provide and the cost structure. NASA will provide more information at a future industry forum and/or in the Draft RFP.

18. Policy and Regulatory Challenges
    a. **What regulatory hurdles remain unresolved (e.g., spectrum allocation, space debris management)?** NASA is still evaluating policy and the regulatory landscape as we draft requirements and will provide more information at a future industry forum and/or in the Draft RFP.

GTA/GFS

19. How will NASA fund the GTAs and RSAAs referenced in the charts? And how will NASA fund the initial set of commercial flights it plans to purchase? NASA funding will be managed via the nominal NASA PPBE process. NASA is evaluating lessons learned from other contracts and the use of GTAs. Details pertaining to GTAs and RSAAs will be provided at an industry forum and/or the Draft RFP but are expected to be similar to past contracts.

20. Could a high-level list of potential NASA-provided Exploration Subsystems be provided? NASA is evaluating GTAs and GFE that may be offered under the contract including Exploration subsystems. A full list will be provided at an industry forum and/or in the Draft RFP.

21. When will and what types/class of NASA equipment/services/exploration subsystems to be offered to CLD providers be available? NASA is evaluating GTAs and GFE that may be offered under the contract including Exploration subsystems. A full list will be provided at an industry forum and/or in the Draft RFP.

22. Slide 6 of the CLD Phase 2 Industry Acquisition Update notes that NASA intends to offer more support/services than previously planned through RSAAs or GTAs. Please explain the types of services/support NASA is considering. NASA is evaluating GTAs and GFE that may be offered under the contract including Exploration subsystems. A full list will be provided in an industry forum and/or in the Draft RFP.

TRANSPORTATION

23. Will crew and cargo services be procured directly from Flight Providers, or could integrators be eligible to aggregate and bid these services? Initially, NASA plans to procure transportation services directly from certified transportation providers for NASA missions, with plans to transition the full responsibility of transportation to the CLD provider at a later date. CLD Providers may propose an alternative approach as part of their proposals if they do not want to utilize NASA-procured transportation. NASA does not plan to procure crew and cargo services for missions that do not have a NASA need, such as an exclusive Private Astronaut Mission (PAM) to the CLD.

24. How will the initial crew and cargo transportation services be procured? Will there be a new acquisition or will existing contracts be utilized? NASA is still evaluating crew and cargo transportation procurement options and will share more information at an industry forum at a later date.

25. Are CLD providers still responsible for flights that launch, outfit, and ready their destination for NASA use? NASA does not currently plan to provide transportation for CLD assembly flights, but may provide, or partner with the CLD contractor(s) to provide cargo transportation for outfitting the CLD, if those outfitting flights include NASA requirements.

26. Do you anticipate also covering crew and cargo needs of NASA's international partners, or NASA's direct needs only? NASA envisions mixed missions including NASA, International Partners (IPs), and other private astronauts. NASA-provided transportation may be utilized for mixed mission needs. A type of value exchange for use of transportation for non-NASA customers is expected. Details would be negotiated as part of the contract.

27. Will/could international contributions be used to fulfill initial NASA crew or cargo transportation services? NASA will procure transportation for initial missions utilizing currently certified vehicles. However, for future flights, NASA may decide to on-ramp new certified vehicles and/or utilize IP-provided vehicles for future missions based on NASA and industry needs.

28. Can you confirm that, during the period NASA procures crew transport services, NASA will be responsible for the full transport and space station cost for the four crew members? Although NASA will provide initial transportation and still expects to be one of many customers on CLDs; the agency will reserve one to two seats per flight for International and/or non-NASA astronauts. The consideration that NASA would receive in exchange for those seats is yet to be determined.

29. Can you provide further clarification/specificity around NASA providing crew/cargo services to the CLD –what does that encompass? How are you defining "missions associated with NASA services"? Anticipated frequency? Can CLDs utilize excess capacity? NASA plans to order full crew and cargo missions and plans to offer any excess capacity to the provider, including for crew and cargo. NASA only plans to provide transportation services for missions related to NASA's needs (i.e., involving NASA crew and payloads) although additional missions could be proposed by the CLD provider.

30. Can you describe the process and timeline through which more details will be released on the scope and duration of these initial transportation services? NASA is still evaluating the scope and duration of the initial transportation services and plans to provide more information at an industry forum and/or the Draft RFP.

31. What is the planned acquisition milestone schedule for the NASA crew/cargo transportation? NASA is still evaluating its crew and cargo acquisition strategy and will share information in this regard as soon as the agency is able to do so.

32. How will NASA's crew and cargo service purchases be distributed among providers? NASA is still evaluating the scope and duration of the initial transportation services and plans to provide more information at an industry forum and/or in the Draft RFP.

33. "Initially, NASA will provide transportation for all missions associated with NASA services (Resupply and Crew)"
    a. **What is NASA's initial assumption on number of missions and transition timeline for the crew and cargo flights?** NASA is still evaluating the scope and duration of the initial transportation services and plans to provide more information at an industry forum and/or in the Draft RFP.
    b. **How will the manifesting for crew and cargo be adjudicated if NASA is the provider of the flights?** NASA plans to order the full crew and cargo missions and plans to offer any excess capacity to the provider, including for crew and cargo. The agency plans to reserve one to two seats per flight for International and/or non-NASA astronauts. NASA is still evaluating manifests and any adjudication processes.
    c. **What role will NASA play as part of the Pre-flight integration and flight operations integration between crew and cargo transportation providers and CLDP providers?** NASA is still evaluating the Government provided services it may provide and plans to provide more information at an industry forum and/or in the Draft RFP.

34. "CLD and transportation providers will be responsible for meeting NASA's certification requirements."
    a. **Is there an example or outline of NASA's certification requirements and process? How does NASA plan to differentiate the certification of short duration mission/crew tended platforms from those that provide a continuous**

presence? NASA's certification requirements and process are still being finalized and will be defined and provided as a part of the Draft RFP. Certification is planned to be similar to the CCP process for crew transportation vehicles. NASA is still evaluating if there are any differences between that process and what will be levied under the CLD Phase 2 Contract.

REQUIREMENTS

35. Does NASA anticipate the insight or oversight requirements for this procurement are going to be managed like the ISS program, or will it be more like the current Commercial Crew program? NASA is still evaluating planned Insight and Oversight Approach and plans to share this information at a future industry forum and/or in the Draft RFP.

36. Is NASA planning to levy the "International Space Power Systems Interoperability Standards (ISPSIS)" and "Modular Electronics Standard for Space Power Systems" for the CLD effort? NASA is still evaluating requirements and finalizing its approach and will share information as the agency is able to at the Safety and Requirements Workshop. https://sam.gov/opp/405df8857dfd451688a40d2514d74819/view

37. Is NASA planning to levy the S-Band Common Communications for Visiting Vehicles (C2V2) as a standard interface protocol for Visiting Vehicles docking with a Commercial LEO Destination? If not, is NASA planning to levy a commercial standard equivalent to C2V2 to eliminate the likelihood of incompatible proprietary protocols between competing Destination OEMs and Visiting Vehicle OEMs? NASA is still evaluating requirements and finalizing its approach associated with standardized communication requirements and plans to share such information at a future industry forum and/or in the Draft RFP.

38. Will structural health monitoring (SHM) and/or condition-based maintenance (CBM) be requirement for the new LEO destination spacecraft, habitats, and structures? If not, why not?" NASA is still evaluating whether formal requirements will be levied in this area or whether these are derived requirements based on provider designs, material, and environment. NASA plans to share more information at a future industry forum and/or in the Draft RFP.

CERTIFICATION/QUALIFICATION

39. How will crew and cargo vehicles be qualified to new station destinations? NASA will work directly with transportation providers to determine the delta qualification that will be needed and develop a forward plan on qualification efforts.

40. How does NASA plan to participate in certification? And will certification be included in the procurement? NASA is evaluating its approach to certification and plans to release more information at an upcoming industry forum and/or in the Draft RFP.

MINIMUM INITIAL OPERATING CAPABILITY (IOC) AND FULL OPERATIONAL CAPABILITIES (FOC)

41. Will there be opportunities and budget for capabilities beyond that date? (“Full capability required by December 2031”) NASA plans to procure on-orbit services from the CLD Provider beyond December 2031 to an end date to be determined. NASA is also interested in any additional capabilities CLD providers may offer in addition to the requirements outlined in the RFP. NASA is considering a CLIN for optional services in the Draft RFP.

42. When does NASA plan to provide the details on Min IOC? (Reference: Slide 6: “Min IOC required by December 2029 (Details of Min IOC still in work)”) NASA is still evaluating the details and definition of Minimum IOC and plans to provide more information at an industry forum and/or in the Draft RFP.

43. What is the definition of “Full capability”? (Reference: Slide 6: “Full capability required by December 2031”) NASA plans to define the initial operating capability (IOC) and full operational capabilities (FOC) for the CLD at an industry forum and/or in the Draft RFP.

44. Does “Full capability required by December 2031” imply that NASA does not expect any additional capabilities to be deployed after December 2031? I.e., must the full set of proposed capabilities be operational by December 2031? NASA plans to require FOC by December 2031 and is evaluating the need for capabilities beyond the ‘full capability requirement set.’ It is anticipated that if any additional capabilities are needed beyond the ‘full operational capability,’ that those will be procured though the provider’s CLD Phase 2 contract.

45. Will NASA define initial and full capabilities for a commercial destination before the draft RFP? In a process for public review similar to this? NASA plans to define the initial operating capability (IOC) and full operational capabilities (FOC) for the CLD at an industry forum and/or in the Draft RFP.

46. “NASA will require a minimum of two NASA astronauts continuously living in LEO by 2031.”
    a. **Is this two per CLD provider or between the two CLD providers?** NASA plans to require continuous human presence by December of 2030 based on having multiple CLD providers, but it is still to be determined if NASA will award none, one, or multiple CLD contracts.

b. **Annually or semi-annually?** NASA is currently planning for 6-month missions during continuous presence but may still evaluate the need for shorter or longer duration missions.

INDUSTRY ENGAGEMENT

47. The slides mention a "Requirements and Safety Workshop in January with release of documents". Will this be an in-person workshop open to all interested parties? What documents do you anticipate releasing? NASA plans to hold this workshop in person at JSC for all interested parties. Documents pertaining to this workshop are publicly available in SAM.gov. https://sam.gov/opp/405df8857dfd451688a40d2514d74819/view

48. When will this Workshop be scheduled? (Reference: Slide 8: "Rqmts & Safety Workshop Jan 2025"). See posting on SAM.gov: https://sam.gov/opp/405df8857dfd451688a40d2514d74819/view

49. When is Industry Day scheduled? Will it be in person/virtual or both? NASA tentatively plans to hold an in-person meeting in March or April 2025 for Industry Day. Virtual options are also being considered.

50. Will one on one meetings be offered as part of Industry Day? NASA tentatively plans to hold an in-person meeting in March or April 2025 for Industry Day. NASA plans to hold one-on-one meetings as part of this Industry Day.

51. How will industry engage in the LMCR activities? NASA plans to continue consulting with internal and external stakeholders to ensure NASA's low Earth orbit (LEO) plans remain aligned with NASA's CLD Phase 2 procurement. NASA intends to hold regular workshops to facilitate this consultation. NASA's LEO Microgravity Strategy is available at this link: https://www.nasa.gov/wp-content/uploads/2024/12/2024-12-lms-final-goals-and-objectives.pdf